Assignment 1: University Course Registration System

Problem Statement

Design a **University Course Management System** in Java.

The university offers different types of courses:

- RegularCourse
- OnlineCourse

Each course has:

- courseId
- courseName
- baseFee

Different course types calculate total fee differently.

Requirements:

- Use **inheritance** to design course types.
- Use **method overloading** for calculateFee().
- Maintain a **static counter** to track total courses created.
- Use an **array of Course objects** (mixed types).
- Demonstrate **constructor chaining** using super().
- Avoid using if-else for course type checking — use polymorphism.

In main:

- Create at least 4 course objects.
- Store them in a Course[] array.
- Print final fee of each course.
- Print total courses created (static).

Assignment 2: Company Payroll System

Problem Statement

Design a **Payroll System** for a company.

There are different types of employees:

- FullTimeEmployee
- PartTimeEmployee
- Intern

Each employee calculates salary differently.

Requirements:

- Create a base class Employee.
- Use **inheritance** for employee types.
- Use **constructor chaining** to initialize employee details.
- Use **method overloading** for calculateSalary().
- Maintain a **static employee counter**.
- Store different employees in an **Employee[] array**.
- Use polymorphism to calculate salary.
- Do NOT use if-else to determine employee type.

In main:

- Create at least 5 employee objects.
- Loop through Employee[] and print salaries.
- Display total employees created.

Assignment 3: Vehicle Toll Management System

Problem Statement

Design a **Toll Management System** for vehicles passing through a toll booth.

Vehicle types:

- Car
- Truck
- Motorcycle

Each type has a different toll calculation rule.

Requirements:

- Create base class Vehicle.
- Use **inheritance** for different vehicle types.
- Implement **method overloading** for calculateToll().
- Maintain:
  - static totalVehiclesProcessed
  - static totalTollCollected
- Use an **array of Vehicle objects**.
- Use **constructor chaining** via super().
- Avoid type checking using if-else; use polymorphism.

In main:

- Create multiple vehicle objects.
- Store them in Vehicle[] array.
- Loop and calculate toll.
- Update static totals.
- Display total toll collected.